Mobile-Based Virtual Physics Laboratory with AI Assistance and Interactive Learning for Rural Students

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Project Proposal Report

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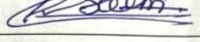
Sri Lanka Institute of Information Technology Sri Lanka

March 2026

DECLARATION

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT (Maximum 300 words)

Rural school students in Sri Lanka face significant barriers to quality physics education, particularly the absence of laboratory facilities and limited access to interactive learning support. Although digital platforms have emerged to address access to experimentation, few systems integrate real-world contextual learning with intelligent student support at the lesson level. Existing virtual physics platforms provide either simulation activities or static quiz assessments independently, without connecting real-world events to curriculum topics or offering an intelligent mechanism for students to resolve conceptual confusion after assessment.

This research proposes a two-part intelligent learning support component within the PhysiV-Lab mobile virtual physics laboratory system, targeting Grade 6-11 students in rural Sri Lanka. The first part is a Real-World News-Based Scenario Question Generator, which automatically collects news articles from curated international sources via RSS, classifies physics-relevant content using zero-shot NLP, maps events to grade-appropriate curriculum topics, and generates contextual scenario-based questions using a locally deployed large language model. The second part is an AI-Powered Post-Lesson Q&A Agent, which activates after lesson MCQ completion and accepts free-text student questions in English or Sinhala, automatically extracts physics keywords using NLP, retrieves grade-filtered explanations via BM25 ranking over a structured physics knowledge base, and falls back to LLM-generated responses when the knowledge base does not contain a match.

Technologies used include Python FastAPI for the backend, feedparser for RSS collection, facebook/bart-large-mnli for zero-shot classification, Ollama with Llama 3.1 for local LLM inference, spaCy and rank_bm25 for keyword extraction and retrieval, Helsinki-NLP models for Sinhala-English translation, and Flutter for mobile interface development. All processing uses open-source tools at zero cost, with the knowledge base and curriculum mapping constructed as original research artefacts from the Sri Lankan physics curriculum.

Expected outcomes include a deployed end-of-lesson scenario question pipeline grounded in current global events, and a responsive multilingual Q&A agent that reduces student conceptual confusion before the scenario challenge. Together these components form a closed real-world contextual learning loop that advances physics education accessibility for rural learners. The system supports SDG 4 (Quality Education) and SDG 10 (Reduced Inequalities).

Keywords: Real-World Contextual Learning, Scenario-Based Assessment, Retrieval-Augmented Generation, BM25 Information Retrieval, Multilingual AI Agent, Mobile Physics Education

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
BM25	Best Match 25 (Information Retrieval Algorithm)
LLM	Large Language Model
NLP	Natural Language Processing
RAG	Retrieval-Augmented Generation
RSS	Really Simple Syndication
SDG	Sustainable Development Goal
Q&A	Question and Answer
MCQ	Multiple Choice Question

Table 1:List_Of_Abbreviations

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1. INTRODUCTION

1.1 Background and Context

Real-World Problem

Physics education in rural Sri Lanka is severely constrained by the absence of laboratory infrastructure, limited instructional support, and a persistent gap between theoretical content and practical application [1], [2]. While several virtual laboratory and mobile learning platforms have been proposed to address access to experimentation, most of these systems confine students to static simulations and predefined quiz assessments that are disconnected from the wider world students inhabit. This creates a conceptual wall between physics as a curriculum subject and physics as the explanation for observable global events.

A particularly underexplored limitation in existing educational technology is the absence of mechanisms that link real-world current events to physics curriculum content in a grade-adapted, automated, and contextually meaningful way. When major events occur globally, such as missile launches, space missions, earthquake disasters, or engineering accidents, these events contain rich physics content that could be leveraged to situate classroom knowledge in lived reality. No existing mobile physics learning platform for school-level students in developing contexts has operationalized this connection.

Simultaneously, students completing lesson assessments frequently encounter conceptual confusion that they cannot resolve independently. Existing systems either provide no post-assessment support, or offer generic AI chatbot interaction that is not grounded in the specific curriculum topic just studied. A need therefore exists for an intelligent, responsive, and multilingual post-lesson support mechanism that operates within the lesson workflow rather than as a separate application.

Who is Affected and Why It Matters

The population most significantly affected includes approximately 1.2 million students enrolled in Grades 6 through 11 in government schools across Sri Lanka, a significant proportion of whom are in rural or semi-urban areas with limited laboratory access [2]. Physics consistently registers among the weakest performance areas in national examinations, with rural students disproportionately affected. The inability to connect theoretical concepts to real-world phenomena is cited by educators as a key contributor to low engagement and retention in physics [1], [3]

Statistical Evidence

Research by Vijayatheepan [2] documents that virtual laboratory access is unavailable in the majority of rural Sri Lankan schools, while Rahman et al. [1] demonstrate that virtual experiments can improve conceptual understanding by up to 34 percent compared to lecture-only instruction. Studies in contextual learning consistently show that grounding abstract scientific content in real-world events improves both motivation and retention, particularly for secondary school learners [4], [6]. Despite this evidence, no mobile platform for Sri Lankan school physics currently implements automated real-world contextual question generation.

Analysis of Existing Solutions and Their Limitations

Existing virtual physics platforms such as PhET Interactive Simulations and Labster provide interactive experiment environments but do not include contextual question generation from live news or post-lesson AI-assisted Q&A [3]. Gamified learning platforms such as Kahoot and Quizziz offer engaging quiz formats but are not curriculum-aligned, do not adapt to grade level, and do not incorporate real-world scenario content. Intelligent tutoring systems and AI chatbots such as Khanmigo provide general subject assistance but are not integrated within a structured lesson workflow, do not support Sinhala, and are not designed for low-resource mobile environments [7], [8].

Most critically, no existing system combines automated news-driven contextual question generation with a post-lesson multilingual retrieval-augmented Q&A agent operating within a single mobile lesson workflow. This dual gap is the focus of the proposed research component

SDG Alignment

This component supports SDG 4 (Quality Education) by improving access to contextually rich and intelligently supported physics learning for underserved rural students. It supports SDG 10 (Reduced Inequalities) by delivering equal access to intelligent educational support regardless of geographic location. The research is aligned with the Technology Integration and Management (TIM) research cluster and the Information Technology (IT) specialization.

1.2 Literature Review

A review of ten recent scholarly sources was conducted to examine the current state of research in virtual laboratory education, contextual and scenario-based learning, intelligent content retrieval, and multilingual AI support in educational systems.

Rahman et al. [1] and Vijayatheepan [2] establish the foundational evidence for virtual laboratory effectiveness in developing-country school contexts, demonstrating measurable improvements in conceptual understanding while identifying persistent infrastructure limitations in rural Sri Lanka. Elmoazen et al. [3] provide a systematic review of learning analytics in virtual laboratory environments and conclude that most existing platforms focus on activity logs and experiment interaction without integrating contextual or scenario-based assessment at the lesson level.

In the domain of contextual and scenario-based learning, Kalogiannakis et al. [4] demonstrate through systematic review that gamification and contextual engagement techniques significantly improve science learning motivation and retention in secondary education. Their analysis shows that connecting abstract physics concepts to observable or current-events-based scenarios produces stronger conceptual transfer than isolated assessment. However, the reviewed systems do not implement automated mechanisms for generating such contexts from live data sources.

For information retrieval in educational systems, Gil et al. [5] propose a smart retrieval system for digital educational content using machine learning and demonstrate that keyword-based retrieval with ranking algorithms can effectively match student queries to relevant educational materials. Their architecture, however, is designed as a standalone system rather than an integrated module within a lesson workflow, and does not support grade-filtered retrieval or multilingual input.

In the area of AI-powered educational support, Huang et al. [6] investigate the effects of AI chatbots on student learning outcomes and motivation, finding significant improvements in student engagement when AI assistance is available during or immediately after learning activities. Zawacki-Richter et al. [7] further confirm through systematic review that AI applications in education are most effective when tightly integrated with structured learning workflows rather than deployed as separate tools. Neither study examines retrieval-augmented generation applied to post-lesson physics Q&A or multilingual support for South Asian school learners.

Lewis et al. [9] introduced the concept of retrieval-augmented generation (RAG), demonstrating that combining a retrieval system with a generative language model substantially improves factual accuracy and relevance compared to generation alone. This approach is directly applicable to the proposed Q&A agent, where BM25 retrieval over a curated knowledge base provides grounded answers before LLM generation is used as a fallback. Robertson and Zaragoza [10] formally established the BM25 ranking algorithm as the standard in probabilistic information retrieval, providing the theoretical basis for the keyword-ranked search mechanism in the proposed system.

Existing commercial and open-source educational products further illustrate the gap. PhET simulations [3] and Labster provide excellent experiment environments but offer no contextual question generation or post-lesson Q&A. Khanmigo and similar AI tutors provide conversational physics support but operate independently of lesson workflows, do not support Sinhala, and require persistent internet connectivity inappropriate for rural deployments [7], [8]. RSS-based news aggregation is well established in information systems research as a reliable, low-cost mechanism for structured content collection [11], while zero-shot text classification using transformer models has been validated as an effective approach for topic filtering without training data [12]

1.3 Research Gap

Despite advances in virtual laboratory education, contextual learning research, and AI-powered educational tools, a clear gap exists in the integration of these capabilities within a single mobile physics lesson workflow for rural developing-country learners.

Most existing physics education platforms do not incorporate real-world current events as sources of assessment content. Where scenario-based questions exist in educational research, they are authored manually and remain static, failing to reflect the currency of global events relevant to physics concepts. No system reviewed automatically extracts physics-relevant content from live news sources, maps it to grade-level curriculum topics, and generates adapted scenario questions through a local LLM pipeline.

Post-assessment support in existing systems is either absent or provided through generic AI chatbots that are not grounded in curriculum content, do not operate within the lesson workflow, and do not support the native languages of rural Sri Lankan learners. The combination of automatic keyword extraction, grade-filtered BM25 retrieval over a structured physics knowledge base, and multilingual support with LLM fallback has not been proposed or evaluated in a mobile physics education context.

Dimension	Gap in Existing Literature	This Study Aims To Investigate
Scenario question source	Questions are static or manually authored; no system generates from live news	Whether automated news-driven scenario question generation is feasible and curriculum-aligned
Grade adaptation	Contextual questions are not adapted to grade level or curriculum mapping	Whether grade-filtered news mapping and LLM prompting produce appropriately levelled scenario questions
Post-lesson support	Post-assessment AI support is separate from lesson workflow and not curriculum-specific	Whether an in-workflow AI Q&A agent improves conceptual clarity before the scenario challenge
Multilingual support	Existing AI tools do not support Sinhala input in school physics contexts	Whether automatic language detection and translation enables effective Q&A for Sinhala-speaking learners
Retrieval grounding	AI chatbots generate answers without curriculum-grounded retrieval	Whether BM25 retrieval over a structured knowledge base improves answer relevance over LLM-only generation
Knowledge base artefact	No structured grade-segmented physics knowledge base exists for Sri Lankan secondary curriculum	Whether a manually curated grade-segmented knowledge base can serve as an effective retrieval corpus

Table 2: Research_Gap

2. **OBJECTIVE**

2.1 Main Objective

To design, implement, and evaluate an intelligent real-world contextual learning support component that automatically generates grade-adaptive physics scenario questions from curated news sources and provides a multilingual AI-powered post-lesson Q&A agent with curriculum-grounded retrieval, within a mobile virtual physics laboratory application for Grade 6-11 students in rural Sri Lanka

2.2 Specific Objectives

1. To design and implement an automated RSS-based news collection pipeline that periodically ingests articles from a defined set of curated international news sources and stores structured article content for downstream processing.

2. To design and evaluate a physics relevance classification mechanism using zero-shot NLP that filters collected news articles and identifies those containing content mappable to Sri Lankan Grade 6-11 physics curriculum topics.

3. To design and implement a curriculum mapping artefact and grade-filtering mechanism that assigns physics-relevant news articles to appropriate grade bands and lesson topics within the national physics syllabus.

4. To implement and evaluate a local LLM-based scenario question generation pipeline that produces grade-adapted, curriculum-aligned multiple choice questions from physics-relevant news content, with structured JSON output stored for delivery.

5. To design and implement an AI-powered post-lesson Q&A agent that accepts free-text student questions in English or Sinhala, extracts physics keywords automatically, retrieves grade-filtered explanations from a structured knowledge base using BM25 ranking, and generates fallback responses using a local LLM when retrieval does not produce a confident match.

6. To construct and evaluate a structured grade-segmented physics knowledge base from Sri Lankan secondary curriculum materials as the primary retrieval corpus for the Q&A agent.

3. METHODOLOGY

3.1 Research Approach

This study adopts a Design Science Research methodology, which is well established in Information Systems research for designing, implementing, and evaluating software artefacts that address identified practical problems [13]. The study will proceed through four phases: problem identification and requirements analysis, system design and artefact construction, prototype implementation, and experimental evaluation. Evaluation will be conducted through functional testing, pipeline output quality assessment, and usability evaluation with student participants where feasible.

3.2 Part One: News-Based Scenario Question Pipeline

3.2.1 News Collection

News articles will be collected quarterly from a defined set of up to fifteen international news sources including BBC, CNN, Al Jazeera, and Reuters, using their publicly available RSS feeds. The feedparser Python library will be used to parse feed content, and article text will be extracted and stored in Cloud Firestore with metadata including source, publication date, and headline. No paid API access is required. The quarterly collection will be triggered manually during the research phase, with a Python script executing the full pipeline on the researcher's workstation.

3.2.2 Physics Relevance Classification

Collected articles will be classified for physics relevance using zero-shot text classification with the facebook/bart-large-mnli transformer model from HuggingFace. A set of candidate physics concept labels aligned to the Sri Lankan curriculum will be defined, including motion and forces, energy and waves, electricity, pressure and fluids, light and optics, heat and temperature, and not related to physics. Articles receiving a top prediction of not related to physics above a confidence threshold of 0.6 will be discarded. Passing articles will be tagged with their predicted physics concept category for downstream mapping.

3.2.3 Curriculum Mapping

A curriculum mapping artefact will be manually constructed as a structured JSON document that maps each physics concept category to grade bands (Grades 6-7, 8-9, and 10-11) and specific lesson topics within the Sri Lankan national physics syllabus. This artefact constitutes an original research contribution. Articles tagged by the relevance classifier will be matched against this mapping to determine their eligible grade bands and associated lesson topics. Articles with no plausible curriculum mapping will be excluded from question generation.

3.2.4 Scenario Question Generation

For each article passing the curriculum mapping stage, a locally deployed Llama 3.1 8B model via Ollama will generate a grade-adapted multiple choice scenario question. A structured

prompt will provide the article context, the target physics concept, the specific lesson topic, and the target grade band, instructing the model to produce a single four-option MCQ with one correct answer and an explanation, formatted as JSON. Questions will be validated for structural completeness before storage in the Firestore scenario_questions collection, tagged with grade range, lesson topic, news source, article date, and physics concept.

3.2.5 Question Delivery

At the end of the post-lesson Q&A session, the Flutter application queries Firestore for a scenario question matching the student's grade and the lesson topic just completed. If multiple matching questions exist, one is selected. If no question exists for a topic-grade combination, a pre-authored fallback scenario question is served. The student receives one scenario question per lesson as the final challenge before lesson completion is recorded.

3.3 Part Two: Post-Lesson AI Q&A Agent

3.3.1 Lesson Trigger and Q&A Session

The Q&A agent activates only after the student has completed all MCQ questions in Wijerathna's assessment module. A free-text input form is then presented with the prompt Do you have any questions about this lesson? The student may submit as many questions as desired. The session is mandatory before the scenario question is unlocked, but the student may proceed by tapping the I am Ready button at any point after at least one question has been asked and answered, or by choosing to proceed directly if they have no questions.

3.3.2 Language Detection and Translation

On submission of each student question, the langdetect Python library will detect whether the input is in Sinhala or English. If Sinhala is detected, the Helsinki-NLP/opus-mt-si-en transformer model will translate the question to English before processing. After the answer is generated, it will be translated back to Sinhala using the Helsinki-NLP/opus-mt-en-si model before display. This ensures full bidirectional Sinhala support using locally deployed models with no API cost or cloud dependency.

3.3.3 Keyword Extraction

The translated or original English question will be processed by spaCy using the en_core_web_sm model to extract candidate physics keyword phrases through noun phrase and named entity detection. Extracted candidates will then be passed to the Ollama LLM with a focused prompt requesting identification of physics-specific keywords from the candidate list, returning a JSON array. This two-stage approach reduces unnecessary LLM calls for straightforward questions while improving extraction accuracy for complex phrasing.

3.3.4 BM25 Retrieval with Grade Filtering

The physics knowledge base stored in Firestore will be filtered by the student's grade level before BM25 ranking is applied. The rank_bm25 Python library will score knowledge base documents against the extracted keyword array. The highest-scoring document will be returned

as the answer if its BM25 score exceeds a defined confidence threshold. Grade filtering ensures that a Grade 6 student asking about force receives a simplified explanation while a Grade 11 student receives a vector-based treatment of the same concept.

3.3.5 LLM Fallback

If BM25 retrieval does not produce a result above the confidence threshold, the student question and extracted keywords will be passed to the Ollama LLM as a physics explanation request with grade-level context in the prompt. The LLM response will be clearly marked as a general explanation rather than curriculum-specific content. This prevents the agent from ever returning a blank response to the student while maintaining transparency about answer provenance.

3.4 Knowledge Base Construction

The physics knowledge base is a manually curated Firestore collection constructed from Sri Lankan Grade 6-11 physics textbooks, teacher guides, and curriculum materials. Each entry covers one physics concept at one grade band, with fields for concept name, grade minimum, grade maximum, plain-language explanation, formula where applicable, and keyword tags. The prototype target is approximately 60 entries covering 20 core concepts across three grade bands. This artefact is an original contribution and will be described and evaluated as such in the research report.

3.5 Evaluation Strategy

Evaluation will be conducted across three dimensions. Pipeline functional testing will verify correct operation of each stage including RSS ingestion, classification, mapping, generation, storage, retrieval, and fallback. Output quality assessment will evaluate generated scenario questions for curriculum alignment, grade appropriateness, factual correctness, and structural completeness using a rubric applied by the researcher and supervisor. Where student participants can be recruited for a pilot session, usability evaluation will assess Q&A agent response relevance and student satisfaction using a short post-session questionnaire. The following benchmarks are intended targets subject to experimental outcomes:

Evaluation Dimension	Planned Metric	Intended Target
Physics relevance classification accuracy	Precision and Recall on held-out article set	Precision >= 0.80, Recall >= 0.75
Scenario question curriculum alignment	Rubric score by supervisor review	>= 80% of questions rated aligned
Grade appropriateness of generated questions	Rubric score by grade band	>= 75% rated appropriate for target grade
BM25 retrieval relevance	Manual relevance assessment on 30 student questions	>= 70% of top results rated relevant
Q&A agent response time	Wall-clock time per query	Response within 8 seconds on development hardware

Sinhala translation accuracy	Human evaluation of 20 Sinhala queries	>= 75% rated accurately translated

Table 3: Planned Evaluation Metrics and Intended Benchmarks

3.6 Ethical Considerations

All news content will be collected from publicly available RSS feeds. No personally identifiable data will be collected from students during evaluation. Student participation in pilot usability testing will be voluntary, and no performance data will be stored beyond anonymized session records for research purposes. The physics knowledge base will be derived entirely from publicly available Sri Lankan national curriculum materials.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

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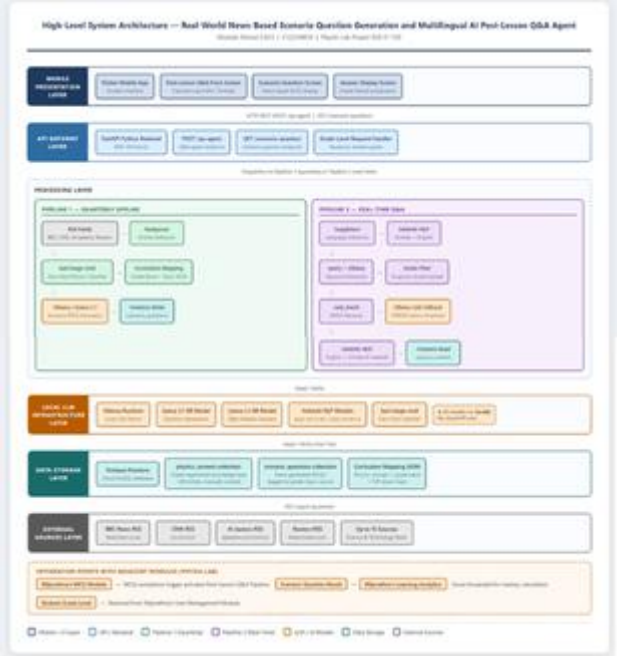


Figure 1:System_Architecture

4.1 Project Structure and Component Scope

The overall PhysiV-Lab project consists of four team-level components: (1) Grade-Adaptive RAG-Based AI Chatbot and Structured Learning Navigation (Sachintha Nirmal S.M), (2) Real-World Scenario Question Generation and Post-Lesson Q&A Agent (this component, Nirmal S.M.S), (3) Adaptive User Management and Learning Performance Evaluation (Wijerathna C.D.W), and (4) Interactive Physics Experimentation and Learning (Vidurangi H.I.D.G). This proposal describes component 2 only.

Within this component, two internal processing pipelines are implemented. The Scenario Question Pipeline operates offline on a quarterly basis and produces a stored question bank in Firestore. The Q&A Agent Pipeline operates in real time during the student lesson session. Both pipelines share the physics knowledge base and Ollama LLM infrastructure.

4.2 Component Position in Lesson Workflow

Within the lesson workflow, this component activates after Wijerathna's MCQ assessment module completes. The Q&A agent session runs first, allowing the student to resolve conceptual confusion from the MCQ phase. Once the student proceeds, the scenario question is served as the final lesson activity. Lesson completion is recorded by Wijerathna's module upon scenario question submission. The component receives the student's grade level and current lesson topic from the system's shared student profile and passes scenario question results back to the performance evaluation module.

4.3 Data Flow

In the quarterly pipeline, RSS feeds are ingested and parsed, article text is classified by the zero-shot model, curriculum-mapped articles are passed to the LLM generator, and generated questions are written to the scenario_questions Firestore collection. In the real-time Q&A pipeline, student question text is received by the FastAPI backend, processed through language detection, translation if required, keyword extraction, and BM25 retrieval or LLM fallback, and the answer text is returned to the Flutter application for display. The Flutter client communicates with the Python FastAPI backend via two endpoints: POST /qa-agent for Q&A queries and GET /scenario-question for lesson-end question retrieval.

4.4 Scalability and Extensibility

The news collection pipeline is extensible to additional RSS sources through configuration only, with no changes to core processing logic. The curriculum mapping artefact can be extended with new topic entries to support additional grade levels or subject areas. The knowledge base can be expanded incrementally. The architecture is modular, with the scenario question pipeline and Q&A agent pipeline operating independently so that either can be modified without affecting the other.

5. USER REQUIREMENTS

5.1 Stakeholder Identification

Stakeholder	Role	Primary Interest
Grade 6-11 Students (Rural)	Primary	Accessible contextual questions and on-demand concept explanations after lesson MCQs
Physics Teachers	Secondary	Curriculum-aligned scenario questions that extend classroom learning with real-world context
System Integration Team	Primary	Correct data formatting and smooth API communication between lesson components
Research Supervisors	Academic	Evidence of novelty, valid methodology, and measurable evaluation outcomes
Educational Authorities	Secondary	Curriculum alignment and potential adoption for national digital learning initiatives

Table 4:Stakeholder_Indentification

5.2 Functional Requirements

ID	Requirement	Priority
FR-01	The system shall collect news articles from up to fifteen configured RSS feed sources on a quarterly schedule.	Must Have
FR-02	The system shall classify each collected article for physics relevance using zero-shot NLP and discard non-relevant articles.	Must Have
FR-03	The system shall map physics-relevant articles to grade bands and lesson topics using the curriculum mapping artefact.	Must Have
FR-04	The system shall generate a structured four-option MCQ scenario question for each mapped article using a locally deployed LLM.	Must Have
FR-05	The system shall store generated scenario questions in Firestore tagged with grade range, lesson topic, source, and date.	Must Have
FR-06	The system shall serve a grade-matched and topic-matched scenario question to the student at the end of each lesson.	Must Have
FR-07	The system shall serve a fallback scenario question when no generated question matches the current lesson and grade.	Must Have
FR-08	The Q&A agent shall activate only after the student has completed the lesson MCQ assessment.	Must Have
FR-09	The Q&A agent shall accept free-text student questions in English or Sinhala.	Must Have
FR-10	The system shall detect question language and translate Sinhala input to English before processing.	Must Have

FR-11	The system shall extract physics keywords from the student question using NLP and LLM-assisted extraction.	Must Have
FR-12	The system shall retrieve the most relevant knowledge base entry for the student's grade using BM25 ranking.	Must Have
FR-13	The system shall generate an LLM-based answer when BM25 retrieval does not exceed the confidence threshold.	Must Have
FR-14	The system shall translate the answer back to Sinhala when the student question was submitted in Sinhala.	Must Have
FR-15	The student shall be able to submit unlimited questions during the Q&A session.	Must Have
FR-16	The Q&A session shall be mandatory before the scenario question is presented to the student.	Should Have

Table 5:Functional_Requirements

5.3 Non-Functional Requirements

ID	Requirement	Intended Target
NFR-01	The Q&A agent shall return an answer within an acceptable response time on development hardware.	Response within 8 seconds per query
NFR-02	The physics relevance classifier shall operate without internet access during classification.	Fully offline local model execution
NFR-03	The LLM inference shall run locally without cloud API dependency.	Ollama local model only
NFR-04	The system shall handle missing or malformed RSS feeds without terminating the collection pipeline.	Zero unhandled pipeline terminations
NFR-05	The knowledge base shall support grade-filtered queries without full collection scans.	Grade-indexed Firestore queries
NFR-06	New RSS sources shall be addable through configuration only without modifying core pipeline logic.	Configuration-only source addition
NFR-07	The Q&A agent shall never return a blank response to the student.	LLM fallback active for all queries

Table 6:Non_Functional_Requirments

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The primary target market is Grade 6-11 students in rural and semi-urban government schools in Sri Lanka who have access to Android smartphones but limited laboratory and instructional support. Secondary markets include private tutoring centers, educational technology providers seeking curriculum-aligned content generation capabilities, and government education authorities interested in scalable digital learning tools for national examination preparation.

6.2 Value Proposition

The proposed component offers two distinct value propositions. For students, it provides the only mobile physics learning tool that connects real global news events to their physics curriculum through automated grade-adapted scenario questions, while also offering on-demand multilingual concept explanation without requiring a separate application or internet search. For educational institutions, it provides a zero-cost, locally deployable pipeline that continuously refreshes physics assessment content from real-world events, reducing reliance on static question banks that quickly become outdated.

6.3 Revenue Model

Should the overall PhysiV-Lab platform be commercialized, this component supports a freemium model in which the core Q&A agent and scenario question delivery are available at no cost, while premium features such as expanded knowledge bases, teacher-facing analytics on frequently asked student questions, and customized news source configurations are offered through institutional subscription. Educational authorities may license the platform for national deployment under a tiered government pricing arrangement.

6.4 Competitive Advantage

No existing mobile physics platform for Sri Lankan school students offers automated real-world news-driven scenario question generation. The multilingual Q&A agent with Sinhala support addresses a language access barrier unmet by internationally developed AI tutoring tools. The fully local processing architecture requires no cloud API subscription, making it suitable for low-resource deployment contexts where recurring API costs are prohibitive.

6.5 Intellectual Property Considerations

The curriculum mapping artefact and grade-segmented physics knowledge base constructed for this research represent original intellectual contributions specific to the Sri Lankan secondary physics curriculum. The news-to-curriculum mapping pipeline architecture may represent a novel software design if validated. Appropriate software copyright and licensing will be established in consultation with the university's commercialization guidelines following academic publication.

7. BUDGET AND JUSTIFICATION

Category	Item	Qty	Unit (LKR)	Total (LKR)
Software	All planned software (open-source: Python, spaCy, HuggingFace, Ollama, feedparser, rank_bm25, Flutter)	—	0	40000
Cloud Services	Firebase Firestore (free tier sufficient for prototype)	—	0	35000
Compute	LLM inference on researcher's personal workstation	—	0	5000
Hardware	Personal laptop for development and offline model inference	—	0	10000
Contingency	Approximately 10 percent buffer for unforeseen expenses	—	—	0
TOTAL				LKR 90000

Table 7:Budget_Summary

Justification

- All software components including Python, spaCy, HuggingFace transformers, Ollama, feedparser, rank_bm25, langdetect, Helsinki-NLP models, FastAPI, and Flutter are have to purchase licenses at RS:30000/=
- LLM inference for both quarterly question generation and real-time Q&A is performed using Ollama on the researcher's existing workstation, eliminating cloud compute expenditure entirely.
- Firebase Firestore's free tier provides sufficient storage capacity for the prototype knowledge base and scenario question collection without incurring service charges.
- The knowledge base and curriculum mapping artefacts are constructed manually from publicly available Sri Lankan national curriculum materials, incurring no data acquisition cost.

8. WORK BREAKDOWN STRUCTURE (WBS)

WBS ID	Planned Activity	Planned Deliverable	Est. Duration
1.0	Requirements and Design		
1.1	Define RSS source list, feed URLs, and collection schema	Source configuration document	3 days
1.2	Construct curriculum mapping artefact for Grades 6-11 physics topics	Curriculum mapping JSON file	1 week
1.3	Define knowledge base schema and target entry list	Knowledge base specification	3 days
1.4	Design Q&A agent API interface contracts with adjacent modules	Interface specification	2 days
2.0	Artefact Construction		
2.1	Populate physics knowledge base from curriculum materials (20 concepts, 3 grade bands)	60-entry knowledge base in Firestore	1.5 weeks
2.2	Construct and validate curriculum mapping table entries	Validated mapping artefact	4 days
3.0	Scenario Question Pipeline Development		
3.1	Implement RSS collection and article storage module	Working collection script	4 days
3.2	Implement and evaluate zero-shot physics relevance classifier	Classifier with evaluation results	1 week
3.3	Implement curriculum mapping and grade filtering module	Working mapping module	4 days
3.4	Implement and evaluate LLM scenario question generation with prompt engineering	Working generation pipeline	1 week
3.5	Integrate pipeline stages and run end-to-end on collected articles	Integrated scenario pipeline	3 days
4.0	Q&A Agent Development		
4.1	Implement language detection and Sinhala translation module	Working translation module	4 days
4.2	Implement keyword extraction using spaCy and Ollama	Working extraction module	4 days
4.3	Implement BM25 retrieval with grade filtering	Working retrieval module	4 days
4.4	Implement LLM fallback module and confidence threshold logic	Working fallback module	3 days
4.5	Implement FastAPI endpoints and integrate all Q&A agent stages	Integrated Q&A agent API	4 days
5.0	Flutter UI Development		
5.1	Implement post-lesson Q&A form screen in Flutter	Working Q&A UI screen	4 days
5.2	Implement scenario question delivery screen in Flutter	Working scenario question screen	3 days
6.0	Testing and Evaluation		

6.1	Pipeline functional testing across all stages	Functional test report	1 week
6.2	Classifier and question quality evaluation using rubric	Quality evaluation report	1 week
6.3	Usability pilot with student participants where feasible	Usability evaluation report	1 week
7.0	Documentation and Reporting		
7.1	Write research report chapter for this component	Draft report chapter	2 weeks
7.2	Prepare presentation materials	Presentation slides	3 days

Table 8:Work_Breakdown_Structure

9. GANTT CHART

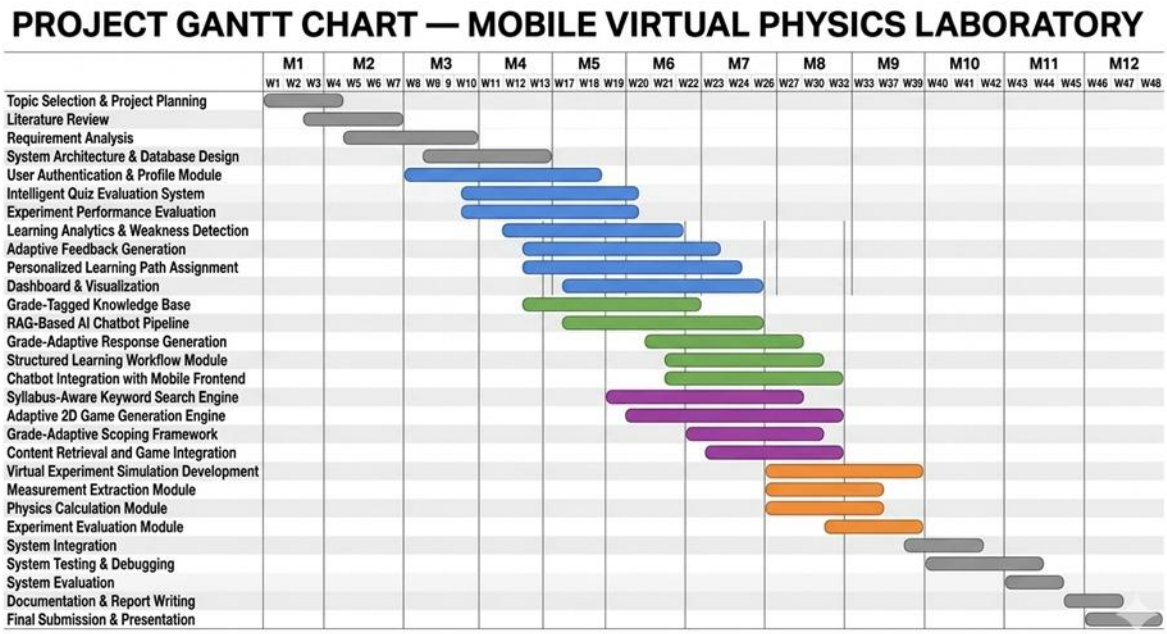


Figure 2:Gantt_chart

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11.APPENDICES

Appendix A: Preliminary Feasibility Study

To assess technical viability before committing to full implementation, a preliminary investigation was conducted into the availability and quality of physics-relevant content in international news RSS feeds. A sample collection of 50 articles was manually gathered from BBC and Al Jazeera RSS feeds and reviewed against the Sri Lankan Grade 6-11 physics curriculum. Preliminary observations indicate that approximately 18 to 25 percent of collected science and technology articles contain content mappable to one or more curriculum topics, with motion and forces, energy, and electricity being the most frequently represented categories. This suggests that a quarterly collection of 200 to 300 articles would yield approximately 40 to 75 physics-relevant articles per cycle, providing a sufficient volume of scenario questions. These are preliminary observations based on manual review only and should not be interpreted as validated classifier performance metrics.

Appendix B: Physics Knowledge Base Sample Entries

The following illustrates the planned structure and content level for knowledge base entries at different grade bands for the concept of velocity:

Concept	Grade Band	Explanation
Velocity	6-7	Velocity is how fast something moves in a specific direction. If a car travels 60 km in one hour heading north, its velocity is 60 km/h north. Velocity is different from speed because direction matters.
Velocity	8-9	Velocity is a vector quantity defined as the rate of change of displacement. It is calculated as v = s/t, where s is displacement in metres and t is time in seconds. Unlike speed, velocity specifies the direction of motion.
Velocity	10-11	Velocity is a vector quantity representing the time derivative of displacement: v = ds/dt. Average velocity is calculated over a time interval, while instantaneous velocity is the limit of displacement over an infinitesimally small time interval. Velocity appears in Newton's Second Law through momentum p = mv.

Table 9:Explanation_01

Appendix C: Risk Analysis

Risk	Likelihood	Impact	Mitigation
RSS feed URL changes or becomes unavailable	Medium	Low	Maintain a list of alternative feeds per source; configuration-only update required
Zero-shot classifier produces low precision on physics articles	Medium	Medium	Tune confidence threshold; supplement with keyword pre-filter based on physics vocabulary list
LLM generates structurally invalid JSON question output	Medium	Medium	Implement output validation with retry logic; maintain fallback hand-authored question set

Ollama model too slow for real-time Q&A on low-spec hardware	Medium	High	Use Phi-3 Mini or Mistral 7B as lighter alternative; cache frequent query responses
Sinhala translation quality insufficient for clear physics explanations	Medium	Medium	Human review of translated outputs during evaluation; fallback to English display with Sinhala label if translation confidence is low
Insufficient pilot participants for usability evaluation	Low	Medium	Conduct evaluation with researcher and supervisor as proxy users; supplement with rubric-based output quality assessment

Table 10:Explanation_02

Appendix D: Curriculum Mapping Table Sample

The following illustrates a sample of the curriculum mapping artefact entries

Physics Concept Tag	Grade Band	Lesson Topic	Example News Event Type
Motion and Forces	6-7	Speed, Distance, Time	Vehicle accidents, sports records, train journeys
Motion and Forces	8-9	Newton's Laws of Motion	Rocket launches, car crashes, sports biomechanics
Motion and Forces	10-11	Momentum and Projectile Motion	Missile trajectories, satellite launches, ballistics
Electricity	8-9	Current, Voltage, Circuits	Power outages, renewable energy projects, electric vehicles
Energy and Waves	6-7	Forms of Energy	Solar panels, wind turbines, earthquake energy
Pressure and Fluids	9-11	Fluid Pressure, Buoyancy	Submarine operations, dam failures, weather pressure systems

Table 11: Explanation_03